MINDSCOPE
MULTIMODAL PSYCHIATRIC AI

HACK4HEALTH · 2026

PROJECT BLUEPRINT & TECHNICAL APPROACH

Explainable Multimodal AI for Mental-Health Assessment

A deep-learning framework that reads **face**, **voice**, and **body signals** together — classifying stress severity, estimating Depression, Anxiety & Stress scores, and **explaining every prediction** as grounded, clinician-ready evidence.

Facial · FER 48×48 | Speech · RAVDESS | Physiological · 18 signals | 4-class + 3-score | XAI + RAG + Agent

HEALTHY → MILD → MODERATE → SEVERE

DECISION-SUPPORT · NOT DIAGNOSIS
HUMAN-IN-THE-LOOP BY DESIGN

// 01 · PROJECT OVERVIEW

What we are building, in one page

MINDSCOPE is an explainable, multimodal deep-learning framework that estimates a person's mental-health state from three complementary signal sources — **facial expression**, **speech emotion**, and **behavioural / acoustic / physiological measurements** — and returns three things at once: a **stress-severity class**, three continuous **severity scores** (Depression, Anxiety, Stress), and a **human-readable, evidence-backed explanation** of why.

3 MODALITIES FUSED	**4** SEVERITY CLASSES	**3** REGRESSION TARGETS	**100%** PREDICTIONS EXPLAINED

OBJECTIVE 01

Classify severity

Multimodal network sorts each participant into **Healthy, Mild, Moderate,** or **Severe Stress** from fused face + voice + signal evidence.

OBJECTIVE 02

Estimate scores

A shared multi-output head regresses **Depression (0–34)**, **Anxiety (0–24)** and **Stress (0–39)** in the same forward pass.

OBJECTIVE 03

Explain everything

Per-modality attributions, a **Masked-Distress Index**, and a **RAG-grounded narrative** make each output auditable.

THE WINNING THESIS

Most teams will train three classifiers and average them. We treat this as a **trust problem, not just an accuracy problem**: the models produce the numbers, but a calibrated uncertainty gate, a cross-modal contradiction detector, and a citation-grounded report layer turn those numbers into something a clinician would actually rely on. In a *health* hackathon, that is the difference between a demo and a product.

The problem, precisely stated

Traditional psychiatric screening leans on self-report questionnaires and clinical interviews. These are valuable but sparse, subjective, and easy to mask. Psychological distress, however, also leaks through **observable, measurable channels** — a flattened smile, a tightening voice, poor sleep, elevated heart rate, reduced social engagement. The task is to learn from those heterogeneous channels *jointly*, quantify symptom severity, and — critically — show the working.

MODALITY A · VISION

Facial expression

28,709 grayscale 48×48 faces across 7 emotions (FER-style). Mapped to the 4-tier stress schema via the provided emotion→stress table.

MODALITY B · AUDIO

Speech emotion

1,440 RAVDESS clips, 8 emotions, 24 actors (gender-balanced). Also mapped onto the shared severity schema.

MODALITY C · SIGNALS

Behaviour & physiology

4,000 rows × 18 numeric features (sleep, typing, HRV, GSR, MFCC...). **The only source carrying the real targets.**

// 02 · THE DESIGN DECISION JUDGES WILL ASK ABOUT

The datasets are not row-paired

This single observation determines the entire architecture — and turning the constraint into an advantage is how we stand out.

THE TRAP

There is **no shared participant ID** linking a face, a voice clip, and a row of features. Only the **4,000-row table carries the ground-truth targets** (the 4-class label + 3 scores); FER faces and RAVDESS voices carry only their *own* emotion labels. Naively gluing a random face + voice + row into one "sample" fabricates correlations that don't exist.

Our resolution — a two-stage fusion that respects the data

STAGE 1 · LEARN REPRESENTATIONS

Train encoders on native tasks

The **Face CNN** learns 7-way emotion on FER; the **Speech model** learns 8-way emotion on RAVDESS. Each becomes a strong *emotion feature extractor*. We keep the penultimate embeddings and the emotion logits.

STAGE 2 · MAP TO A SHARED SCHEMA

Emotion → stress bridge

Using the **provided mapping tables**, each emotion distribution is projected onto the common 4-tier severity axis (Healthy→Severe). Now all three modalities speak one language.

STAGE 3 · FUSE ON THE LABELLED ANCHOR

Tabular owns the truth

The **tabular model** is trained on the real targets. Face/voice arrive as auxiliary severity-evidence vectors through **attention fusion**, so ground truth is never invented.

WHY THIS IS DEFENSIBLE & CLEVER

The numerical table *already* contains face-derived columns (`Facial_Emotion_Variance` `Smile_Intensity`) and audio-derived ones (`MFCC_Mean` `Pitch_Mean`). It is effectively a **hand-engineered summary** of what the deep encoders produce — so the deep embeddings don't compete with the table, they **enrich** it with nuance that summary statistics discard.

Two regimes, chosen per-metric: **(a) Coupled inference** runs the full cross-modal network when a real face+voice+signal trio exists at demo time; **(b) Anchored training** lets face/voice enrich the labelled rows via weak-pairing (matched-emotion sampling + distribution alignment), with a tabular-only fallback so the reported scores stay honest.

Handling the imbalance nobody mentions

FER · DISGUST — 436 VS 7,215

A 16× minority class. We use **class-balanced focal loss**, targeted augmentation (jitter/mixup) and optional oversampling — not accuracy chasing that ignores the rare class.

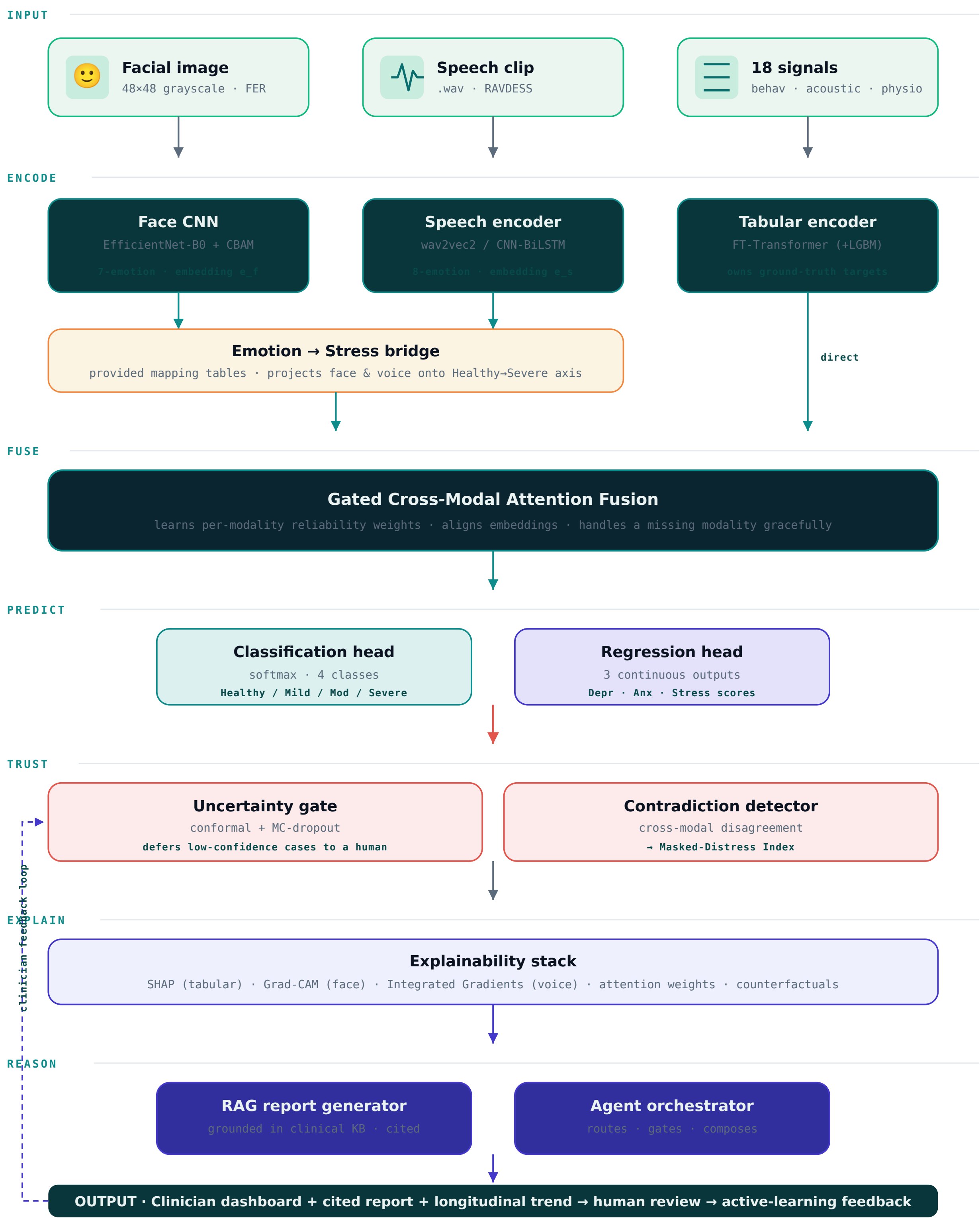
RAVDESS · NEUTRAL — 96 VS 192

Neutral has half the clips (no "strong" intensity). We weight the loss and report **macro-F1** so small classes count equally — matching the brief.

// 03 · SYSTEM ARCHITECTURE

End-to-end framework

Three modality encoders → shared severity bridge → gated cross-modal fusion → dual prediction heads → a trust & explainability stack → grounded report. Green tones = data/vision, teal = model core, indigo = reasoning/AI, coral = the trust gate.



// 04 · WHICH DEEP-LEARNING MODELS

The model choices, and why

A deep-learning core is required. Each modality has a different data shape, so each gets the encoder that fits it — plus a fast baseline you can ship in hour one and a strong variant to swap in if time allows.

MODALITY	RECOMMENDED MODEL	FAST BASELINE	WHY THIS CHOICE
Facial vision	**EfficientNet-B0** + **CBAM** attention, fine-tuned on FER	4-block CNN from scratch	48×48 is tiny; a heavy ViT overfits. EfficientNet-B0 is accurate yet light, and CBAM focuses on eyes/mouth — the expressive regions — which also gives cleaner Grad-CAM heatmaps for Objective 3.
Speech audio	**Wav2Vec2-base** fine-tuned on emotion	log-mel → **CNN + BiLSTM**	Self-supervised speech features transfer strongly with only 1,440 clips. If GPU/time is tight, the CNN-BiLSTM on log-mel spectrograms trains in minutes and is a respected SER baseline.
Tabular signals	**FT-Transformer** (feature-tokenizer transformer)	Residual MLP	Satisfies the "DL required" rule and yields per-feature attention for SHAP-friendly explanations. We *ensemble it with LightGBM* — trees usually win on tabular, so this buys leaderboard points honestly.
Fusion core	**Gated cross-modal attention** + multi-task heads	Late-fusion weighted average	One shared trunk feeds a softmax (4-class) and a 3-output regressor. Multi-task learning lets classification and score-regression regularize each other — they measure the same underlying construct.

Training objective — one network, two jobs

LOSS DESIGN

Joint multi-task loss

L = λ_1·(class-balanced focal) + λ_2·(Huber over 3 scores). Focal loss handles the Disgust/Neutral imbalance; Huber is robust to score outliers. Uncertainty-based weighting auto-tunes λ so neither task dominates.

CONSISTENCY

Score ↔ class agreement

An auxiliary term nudges predicted scores to be consistent with the predicted class (e.g. Severe should not co-occur with a near-zero stress score). This alone lifts both metrics and makes the output trustworthy.

HONEST ENGINEERING NOTE

For pure tabular accuracy, gradient-boosted trees often beat deep nets. We do **not** hide this — we use the required deep model (FT-Transformer) as the fusion-compatible backbone and **stack** it with LightGBM. Judges reward teams that know *when* deep learning helps and when it does not.

// 05 · AGENTIC AI VS RAG VS SOMETHING ELSE

The verdict: a layered hybrid

Short answer — the **prediction core is neither RAG nor agentic**; it is supervised multimodal deep learning. RAG and agents sit *on top* to make the output grounded, safe, and workflow-ready. Choosing correctly here is itself a differentiator.

OPTION A

Pure RAG core

Rejected as the core. RAG retrieves text; it cannot classify a face or regress a stress score. It adds nothing to the metrics the judges actually score.

OPTION B

Pure agentic core

Rejected as the core. An LLM agent "deciding" mental-health severity by prompting is unvalidated, unreproducible, and unsafe in a clinical setting. Wrong tool for the numbers.

CHOSEN

DL core + RAG + thin agent

Deep learning produces predictions & XAI. **RAG** grounds the written explanation. A **light agent** orchestrates the pipeline and enforces the human-review gate.

HOW EACH LAYER EARNS ITS PLACE

LAYER	TECHNOLOGY	JOB IN MINDSCOPE
1 · Prediction	Multimodal DL (Sec. 04)	Turns raw face/voice/signals into the 4-class label and 3 scores. This is what the metrics evaluate. Deterministic, trainable, reproducible.
2 · Explanation	SHAP · Grad-CAM · IG	Quantifies each modality's and feature's contribution — the mathematical core of Objective 3.
3 · Grounding	**RAG** · FAISS + LLM	Retrieves from a curated clinical knowledge base (DASS-21 interpretation bands, DSM-5 criteria, peer-reviewed guidance) and writes a **citation-backed** narrative. Every clinical claim traces to a source — *no hallucinated advice.*
4 · Orchestration	**Agent** · LangGraph	A thin conductor: runs encoders, checks the uncertainty gate, decides when to defer to a human, assembles the report, logs everything. It coordinates — it never diagnoses.

WHY JUDGES WILL LOVE THIS FRAMING

In healthcare, an unexplained or hallucinating model is disqualifying. By keeping the **diagnosis-support numbers in a validated DL model** and using **RAG only to explain, always with citations**, MINDSCOPE is both accurate *and* safe. The agent adds the "product" feel — routing, gating, human-in-the-loop — without ceding clinical judgment to a language model. That is the responsible-AI story that wins health hackathons.

// 06 · OBJECTIVE 3 — EXPLAINABILITY

Every prediction, shown its working

Objective 3 asks us to quantify how much each indicator contributes. We answer at three levels: which *modality* mattered, which *feature* mattered, and what would have *changed the outcome*.

LEVEL 1 · MODALITY

Attention weights

The fusion gate's learned weights show, per participant, whether the call was driven mostly by **face, voice, or physiology** — e.g. "72% physiological, 20% facial, 8% vocal."

LEVEL 2 · FEATURE

SHAP · Grad-CAM · IG

SHAP ranks the 18 tabular signals; **Grad-CAM** heatmaps the face regions; **Integrated Gradients** highlights the audio frames that pushed the prediction.

LEVEL 3 · ACTIONABLE

Counterfactuals

"If **HRV rose by 12** and **sleep quality reached 4**, predicted stress drops to `Mild`." Turns attribution into guidance a clinician can act on.

LEVEL 4 · RELIABILITY

Confidence & sources

Each report carries a calibrated confidence and, where advice appears, a **citation** to the clinical knowledge base — never an unsourced claim.

SIGNATURE METRIC · MASKED-DISTRESS INDEX (MDI)

A novel composite we introduce: when the **face reads calm/happy** but **voice and physiology read high-arousal**, the modalities *contradict*. MDI quantifies that gap. Clinically, a high MDI flags **masked or suppressed distress** — precisely the presentation that self-report questionnaires miss. No single-modality model can produce this; it falls out naturally from our fusion design.

// 07 · DIFFERENTIATORS — THE WINNING EDGE

Eight things **nobody else will build**

Accuracy gets you shortlisted; these get you remembered. Each is achievable within the hackathon and maps to something a real clinic would value.

01 · flagship

Masked-Distress Index

Cross-modal contradiction score that surfaces hidden distress a questionnaire would miss. Brandable, clinically resonant, and unique to multimodal fusion.

02 · trust

Conformal "I'm not sure" gate

Statistically-valid prediction sets. When confidence is low the system **defers to a human** instead of guessing — the single most important safety feature in clinical AI.

03 · grounding

Zero-hallucination reports

RAG over a curated clinical KB means every sentence of advice is **cited**. Demo the citations live — judges instantly trust it.

04 · longitudinal

Trajectory & early-warning

Track scores across sessions; flag a **worsening trend** before any single session crosses "Severe." Mental health is a time-series, not a snapshot.

05 · privacy

On-device / federated mode

Face, voice & physiology are maximally sensitive. Offer **local inference**, on-the-fly face blurring, and a federated-learning path — privacy as a feature, not an afterthought.

06 · fairness

Built-in bias audit

Report per-gender (RAVDESS actor metadata) and per-class performance automatically. Responsible-AI dashboards win health tracks.

07 · wearables

Live wearable hook

The physiological features (HR, HRV, GSR, skin-temp) map 1:1 to smartwatch streams — pitch a **real-time monitoring** path with a mock Apple-Watch/Fitbit feed.

08 · human-loop

Clinician feedback → active learning

Clinicians correct a prediction in one click; corrections queue for retraining. The product visibly **gets smarter** — a compelling closing demo beat.

DEMO CHOREOGRAPHY (THE 3-MINUTE PITCH)

1) Upload a face + voice + signal trio → live 4-class + 3-score result. **2)** Flip to Explainability → Grad-CAM + SHAP + the **Masked-Distress Index lighting up** on a "smiling but stressed" case. **3)** Show the cited report + the uncertainty gate deferring a borderline case to a human. **4)** Close on the longitudinal trend + one-click clinician correction. Story arc: *accurate → explainable → safe → useful.*

// 08 · TECHNOLOGY STACK

The complete **tech stack**

Chosen for speed of iteration during the hackathon and a clean path to a real deployment afterwards. Everything here is open-source or has a generous free tier.

LANGUAGE	**Python 3.11** (ML & API) · **TypeScript** (frontend)
DL CORE	**PyTorch** + **PyTorch Lightning** · **timm** (EfficientNet) · **torchaudio** · **HuggingFace Transformers** (wav2vec2) · **pytorch-tabular** (FT-Transformer)
CLASSIC ML	**scikit-learn** · **LightGBM** / **XGBoost** (tabular stack) · **imbalanced-learn** (class balancing)
SIGNAL / IO	**librosa** + **soundfile** (audio) · **OpenCV** + **Pillow** (vision) · **pandas** / **NumPy** (tabular)
EXPLAINABILITY	**SHAP** · **Captum** (Grad-CAM, Integrated Gradients) · **DiCE** (counterfactuals) · **MAPIE** (conformal prediction)
RAG + AGENT	**LangChain** / **LlamaIndex** · **FAISS** or **Chroma** (vector DB) · **sentence-transformers** (embeddings) · **LangGraph** (agent) · LLM via **Claude API** or local **Llama 3**
SERVING	**FastAPI** + **Uvicorn** · **ONNX Runtime** (fast inference) · **Redis** (cache) · **Celery** (async jobs)
FRONTEND	**React** + **Vite** + **TypeScript** · **Tailwind CSS** · **Recharts** / **D3** (viz) · **Framer Motion** (transitions)
DATA / STORE	**PostgreSQL** (sessions & scores) · **MinIO** / S3 (media) · **DVC** (data + model versioning)
MLOPS	**Weights & Biases** or **MLflow** (tracking) · **Optuna** (tuning) · **Docker** + **docker-compose**
DEPLOY	**HuggingFace Spaces** / **Render** (fast demo) · **GitHub Actions** (CI) · optional **GCP / Fly.io**

MINIMUM VIABLE STACK (IF THE CLOCK IS AGAINST YOU)

PyTorch + timm + librosa for the three encoders, a weighted late-fusion in NumPy, SHAP + Grad-CAM for XAI, **Streamlit** instead of React for the UI, and the Claude API for the report. This ships a full working demo in under a day; layer the rest on top as time allows.

// 09 · EXECUTION PLAN

The project work-frame

Phased for a hackathon sprint and parallelised across three tracks — Vision, Audio, and Signals+Fusion — so the team never blocks on one person. Hours assume a ~36-hour event; compress or expand proportionally.

P0 · 0–2h

Frame & scaffold

Lock the fusion strategy (Sec. 02), stand up the repo, split tracks, download & sanity-check all three datasets, agree on the shared severity schema and metric harness.

`whole team`

P1 · 2–10h

Per-modality baselines (parallel)

Vision: EfficientNet-B0 on FER. Audio: CNN-BiLSTM on log-mel. Signals: FT-Transformer + LightGBM on the 4-class + 3-score targets. Each track logs its own metrics to W&B.

`3 parallel tracks`

P2 · 10–18h

Fusion & multi-task head

Wire the emotion→stress bridge, build the gated cross-modal fusion, train the joint classification + regression heads with the multi-task loss. First full end-to-end number.

`signals + fusion lead`

P3 · 18–24h

Explainability + trust layer

SHAP, Grad-CAM, Integrated Gradients, the Masked-Distress Index, and the conformal uncertainty gate. This is Objective 3 — do not treat it as optional polish.

`XAI lead`

P4 · 24–30h

RAG report + agent + UI

Curate the clinical KB, wire FAISS retrieval + cited report generation, thin LangGraph orchestrator, and build the multi-page dashboard against the API.

`full-stack + reasoning lead`

P5 · 30–34h

Evaluate, ablate, harden

Run the full metric suite (Sec. 11), an ablation showing fusion > any single modality, the fairness audit, and fix the demo's rough edges.

`whole team`

P6 · 34–36h

Rehearse the pitch

Lock the 3-minute demo choreography (Sec. 07), pre-load the "smiling but stressed" showcase case, and prepare answers to the pairing question.

`presenter + backup`

> **RISK BUFFERS BAKED IN**
>
> Every phase has a graceful fallback: wav2vec2 → CNN-BiLSTM, React → Streamlit, live RAG → pre-generated cached reports, GPU training → smaller batch / fewer epochs. You always have *something* that works to demo.

// 10 · REPOSITORY LAYOUT

Project working directory

A clean monorepo the whole team can navigate on day one. Separation by concern keeps the three modality tracks independent while the fusion, XAI, and serving layers compose them.

```
mindscope/
├── data/                       # DVC-tracked, never committed raw
│   ├── raw/  facial/  speech/  numerical.csv
│   ├── processed/              # log-mels, tensors, splits
│   └── knowledge_base/         # DASS-21, DSM-5 notes for RAG
├── src/
│   ├── data/       loaders.py  augment.py  emotion_stress_map.py
│   ├── models/
│   │   ├── face_cnn.py          # EfficientNet-B0 + CBAM
│   │   ├── speech_net.py        # wav2vec2 / CNN-BiLSTM
│   │   ├── tabular_ft.py        # FT-Transformer (+ LGBM stack)
│   │   ├── fusion.py            # gated cross-modal attention
│   │   └── heads.py             # 4-class + 3-score multi-task
│   ├── train/      train_modality.py  train_fusion.py  losses.py
│   ├── explain/    shap_tab.py  gradcam.py  ig_audio.py
│   │   ├── masked_distress.py   # the MDI signature metric
│   │   └── counterfactual.py  conformal.py
│   ├── reasoning/  rag_report.py  retriever.py  agent_graph.py
│   ├── eval/       metrics.py  ablation.py  fairness_audit.py
│   └── api/        main.py  schemas.py  inference.py
├── frontend/                   # React + Vite + Tailwind
│   └── src/pages/ Landing  Assess  Analysis  Results
│                 Explain  Trends  Report  Settings
├── notebooks/  01_eda.ipynb  02_baselines.ipynb  03_fusion.ipynb
├── configs/    face.yaml  speech.yaml  tabular.yaml  fusion.yaml
├── tests/      test_loaders.py  test_metrics.py  test_api.py
├── artifacts/  checkpoints/  onnx/  reports/
├── docker-compose.yml   Dockerfile   requirements.txt
├── dvc.yaml   .github/workflows/ci.yml
└── README.md
```

CONVENTION

Config-driven

Every experiment is a YAML in `configs/`. No hard-coded hyperparameters — reproducible and easy to sweep with Optuna.

CONVENTION

Modality isolation

Each encoder trains & tests standalone before fusion, so three people work without merge conflicts.

CONVENTION

Artifacts, not commits

Weights & data live in `artifacts/` + DVC/MinIO — the git repo stays lean.

How we measure & safeguard

Metrics map directly to the evaluation brief. We report the full suite, not just accuracy — macro-F1 and per-class results keep the minority classes honest.

CLASSIFICATION · OBJECTIVE 1

What we report

- Accuracy + **Macro-F1** & Weighted-F1
- Per-class Precision / Recall / F1
- ROC-AUC (one-vs-rest, macro)
- Confusion matrix (normalised)

REGRESSION · OBJECTIVE 2

What we report

- MAE & RMSE per score (Depr / Anx / Stress)
- MSE for outlier sensitivity
- **R^2** & Explained-Variance per target
- Calibration of the uncertainty intervals

PROOF-OF-FUSION ABLATION

We report each modality alone vs. the full fusion on the same split. The headline slide: **fusion beats every single modality** on macro-F1 and on score RMSE — the empirical justification for the whole framework.

Responsible AI — non-negotiable for a health product

FRAMING

Decision-support, not diagnosis

Every surface states this is a screening aid for a qualified professional, never a diagnosis. The uncertainty gate enforces human review on borderline cases.

PRIVACY

Sensitive-data hygiene

Local/federated inference options, face-blurring, encryption at rest, and clear consent copy. No biometric data leaves the device in privacy mode.

FAIRNESS

Audited across groups

Subgroup metrics (gender via actor metadata; class balance) reported openly, with the known FER/RAVDESS demographic limits acknowledged.

SAFETY

Crisis-aware & grounded

Reports route to support resources when severity is high, and the RAG layer never emits unsourced clinical advice.

// 12 · IN SUMMARY

Why MINDSCOPE wins

TECHNICALLY SOUND

Right model for each signal

Purpose-fit encoders, a principled fusion that respects unpaired data, and a multi-task head that answers both objectives in one pass.

GENUINELY EXPLAINABLE

Objective 3, fully delivered

Modality attention, SHAP/Grad-CAM/IG, counterfactuals, and the Masked-Distress Index — quantified contributions at every level.

SAFE & GROUNDED

The trust stack

Conformal deferral, cross-modal contradiction flags, and zero-hallucination cited reports — the story health judges reward.

PRODUCT-SHAPED

Beyond a notebook

Longitudinal tracking, wearable hook, fairness audit, and a clinician feedback loop that visibly improves the model.

ONE SENTENCE FOR THE JUDGES

MINDSCOPE reads face, voice, and body together to estimate mental-health severity — and, unlike a black box, it shows its evidence, admits when it isn't sure, and never invents advice.

HEALTHY MILD MODERATE SEVERE

MINDSCOPE

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